STAT 351: Final Project Proposal

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1. You and Your Group

My name is Luke Lowther, and I will be working with Ari Crumley.

2. Research Question

My partner and I want to look into a the combination of short rest and playing in high elevation arenas affect NBA teams. These are the questions we would like to research and hopefully answer.

Primary Question:

Is the margin penalty for visiting teams on short rest (back to back or 3 in 4 nights) larger when playing in high altitude arenas (Primarily Denver and Utah) than in lower altitude arenas. We are aiming to investigate whether already fatigued teams struggle more at elevation.

Secondary Questions:

Does the visitor penalty scale with the altitude difference between the visitors own home arena and the game's location?

Is Denver's home advantage larger then other teams? Does it depend / how much does it depend on visitor's rest?

Does the effect show up more in the second half / fourth quarter when the visiting team has been running more at elevation?

Shape of Answer:

We aim to calculate an estimated interaction coefficient (points) with a confidence interval, a comparison of group means, and maybe a fitted model to predict team performance coming into high elevation arenas.

3. Data

Source. I will use the hoopR R package (https://github.com/sportsdataverse/hoopR). Each row represents one team's performance in a game. I will use approximately 22,000 rows from the 2015–2025 NBA seasons, excluding 2019–20. Variables include game scores, dates, teams, offensive and defensive ratings, pace, altitude, travel distance, and rest days.

How you will get it. I will download the data through the `hoopR` package in RStudio. I have already installed the package and successfully loaded NBA game data. I still need to collect altitude and travel information.

If it falls through. If altitude data is too difficult to collect, we will focus only on how short rest affects NBA team performance. We will also use Basketball Reference as a backup source for game data.

4. Planned Analysis

We will start by building a dataset in R, where each row in one NBA regular season game (spanning many seasons, we are thinking the previous 10 excluding 2019-2020 and maybe 2020-2021). For each game, we will calculate how many days of rest each team had, flag it if they are in the second game of a back to back, and record the altitude of the arena. Then we can compute simple statistics like the average point margin for the visiting team, and split it by venue sorted by elevation and rest. That will give us a look at whether tired visitors do worse in high elevation environments rather then at sea level.

To answer our main question, we will fit a linear regression of the visiting team's average point margin on team strength, a back to back indicator (second game), a high altitude indicator, and the interaction between those last two. The interaction coefficient is the key number, because it tells us how many extra points a tired visitor loses in the high elevation. Because Denver and Utah only host a 41 home games a season, and few are the second night of a back to back for the visitors, we are thinking of incorporating all arenas and accounting for elevation. If time allows, we will also test whether the size of altitude change between a visitors home arena and the game arena matters.

As an additional goal, we will fit our regression model on earlier seasons, and see if it can predict margins in the most recent season (2025-2026). We expect the report to be carried by two or three figures, a plot of average margin by venue and rest group with confidence intervals, a coefficient plot for the interaction effect, and a plot of predicted vs actual margins..

5. Timeline and References

Week	Dates	What you expect to have done
Week 10	Oct. 26–30	Data scraped and cleaned (two rows -> one row per game, only relevant columns, add back to back indicator)
Week 11	Nov. 2–6	First statistics calculated, begin making plots
Week 12	Nov. 9–13	Start on conclusion / analysis
Week 13	Nov. 16–20	Exam Nov. 16 Exam week / break for study time
—	Nov. 23–27	Fall break Fall break / catch up work if necessary
Week 14	Nov. 30–Dec. 4	Presentations begin Begin building slides and presentation
Week 15	Dec. 7–11	Report due Dec. 13 Finish presentation and throughough proofread. Turn in Project

References:

https://github.com/sportsdataverse/hoopR

https://www.basketball-reference.com/

6. AI Acknowledgment

We used Claude Sonnet 5.5 to help us brainstorm project ideas. We also used it to help us fully flush out the idea that we chose. At no point was AI used to do any work for us, simply to aid us in coming up with a project idea that was well thought out and somewhat niche (We did not want to accidentally do the same thing as another group).